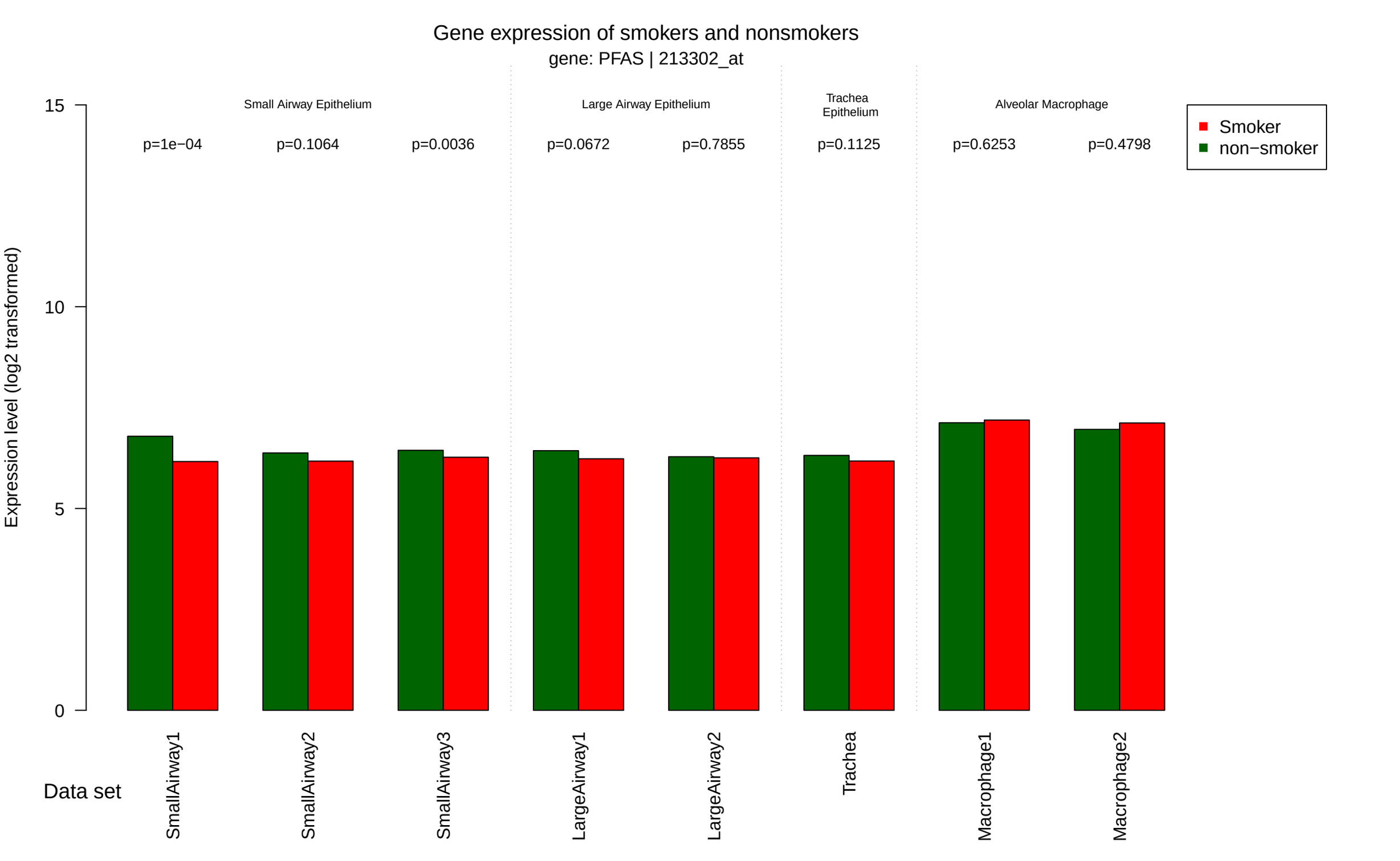
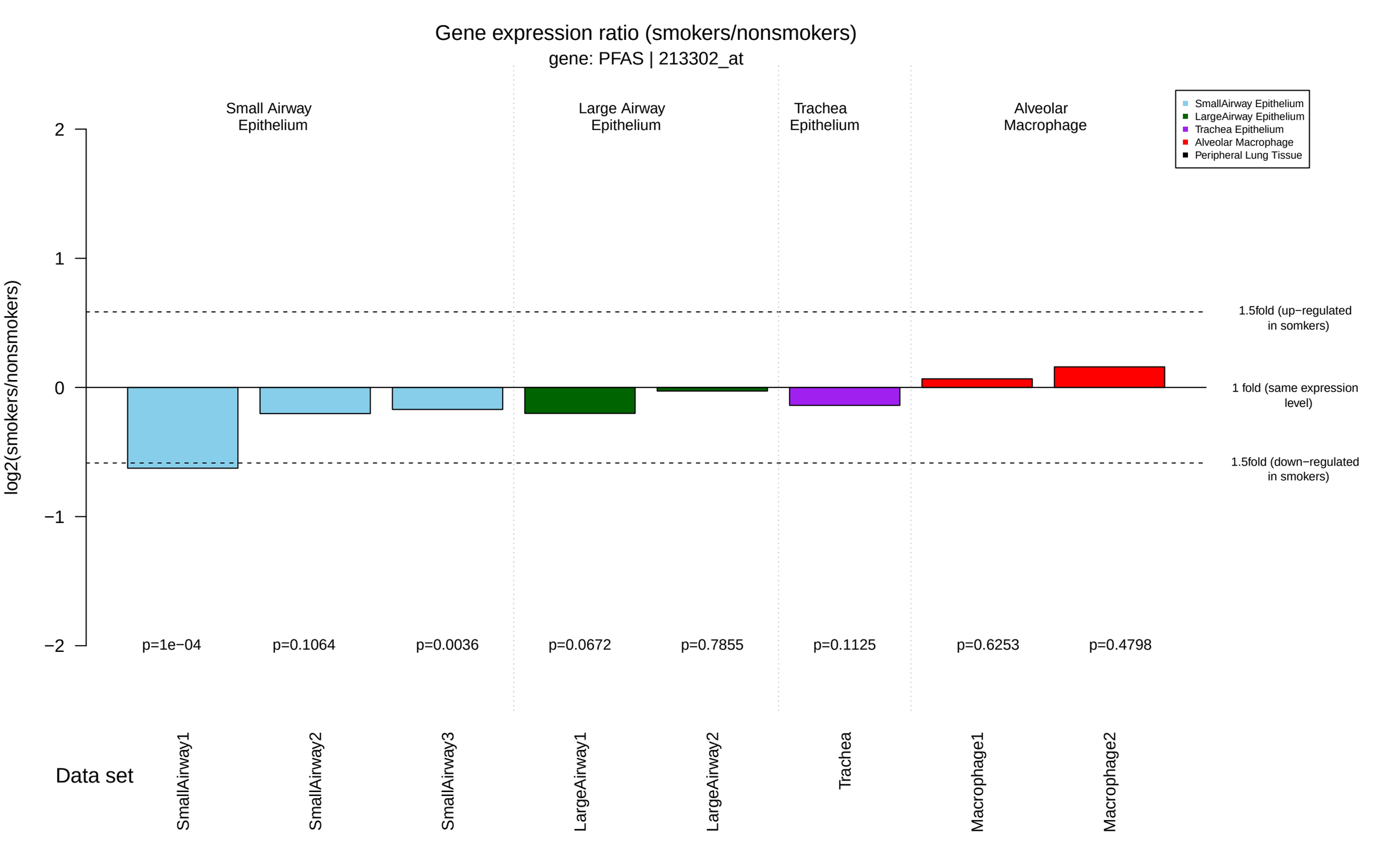
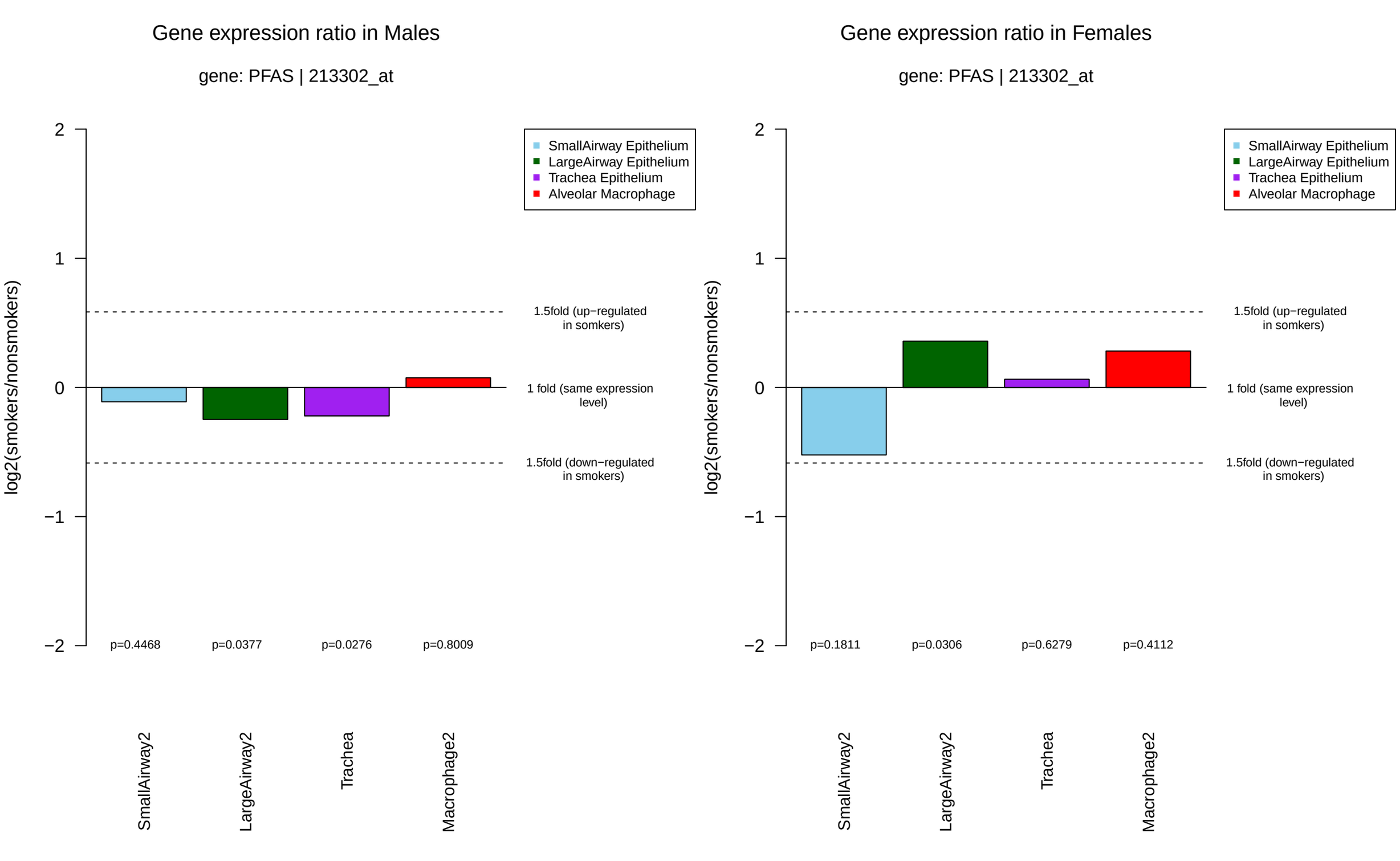
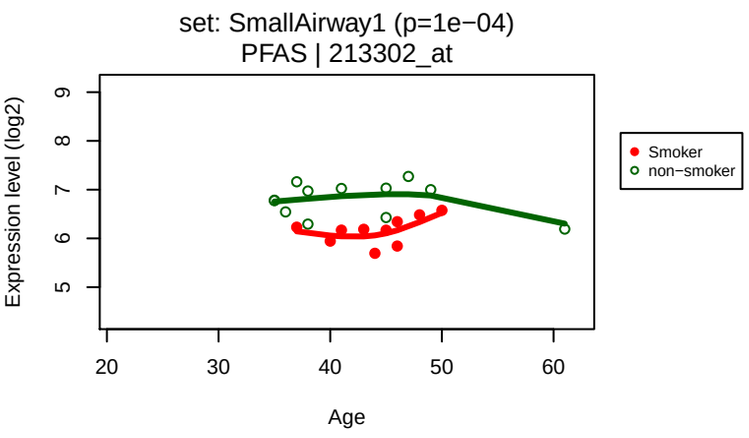
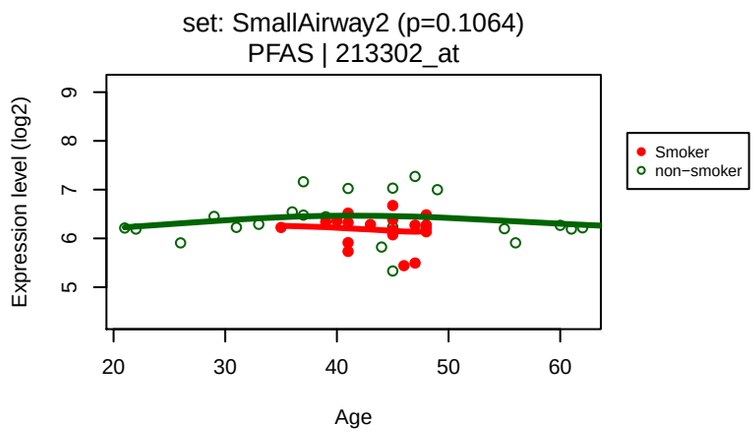
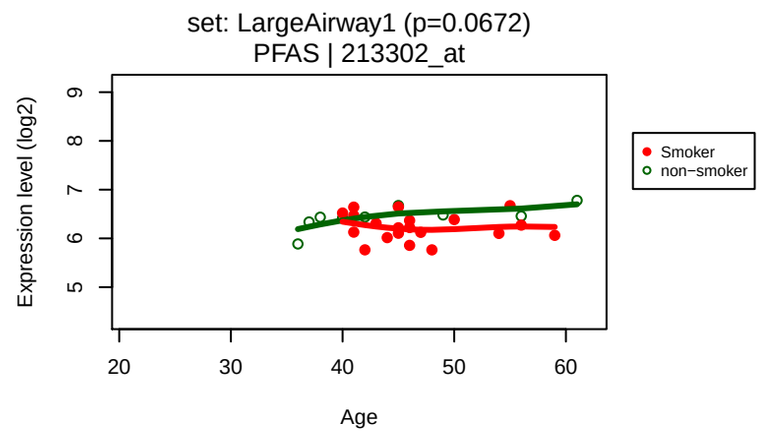

Gene expression of smokers and nonsmokers
gene: PFAS | 213302_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=1e−04
p=0.1064
p=0.0036
p=0.0672
p=0.7855
p=0.1125
p=0.6253
p=0.4798
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

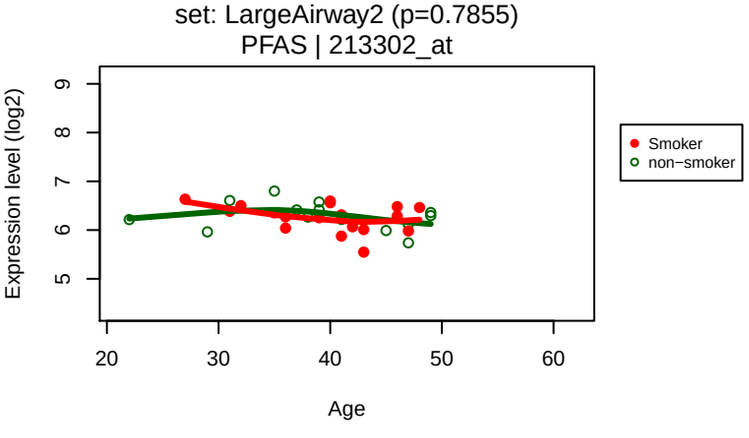

Gene expression ratio (smokers/nonsmokers)
gene: PFAS | 213302_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.1064
p=0.0036
p=0.0672
p=0.7855
p=0.1125
p=0.6253
p=0.4798
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

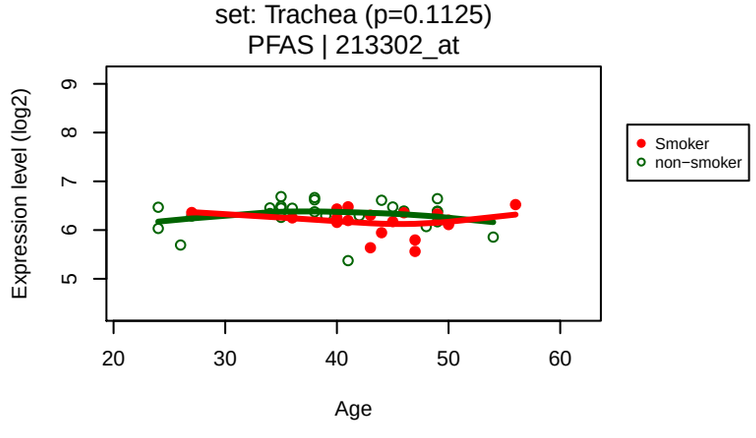

Gene expression ratio in Males
gene: PFAS | 213302_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4468
p=0.0377
p=0.0276
p=0.8009
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PFAS | 213302_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1811
p=0.0306
p=0.6279
p=0.4112
SmallAirway2
LargeAirway2
Trachea
Macrophage2

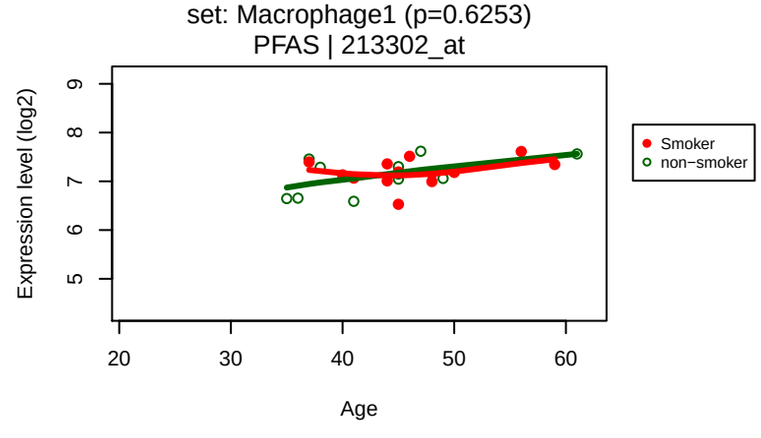

set: SmallAirway1 (p=1e−04)
PFAS | 213302_at
Expression level (log2)
Age
Smoker
non−smoker

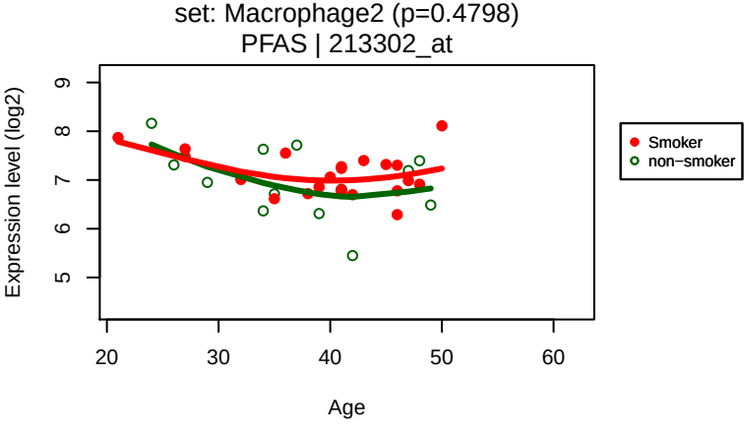

set: SmallAirway2 (p=0.1064)
PFAS | 213302_at
Expression level (log2)
Age
Smoker
non−smoker

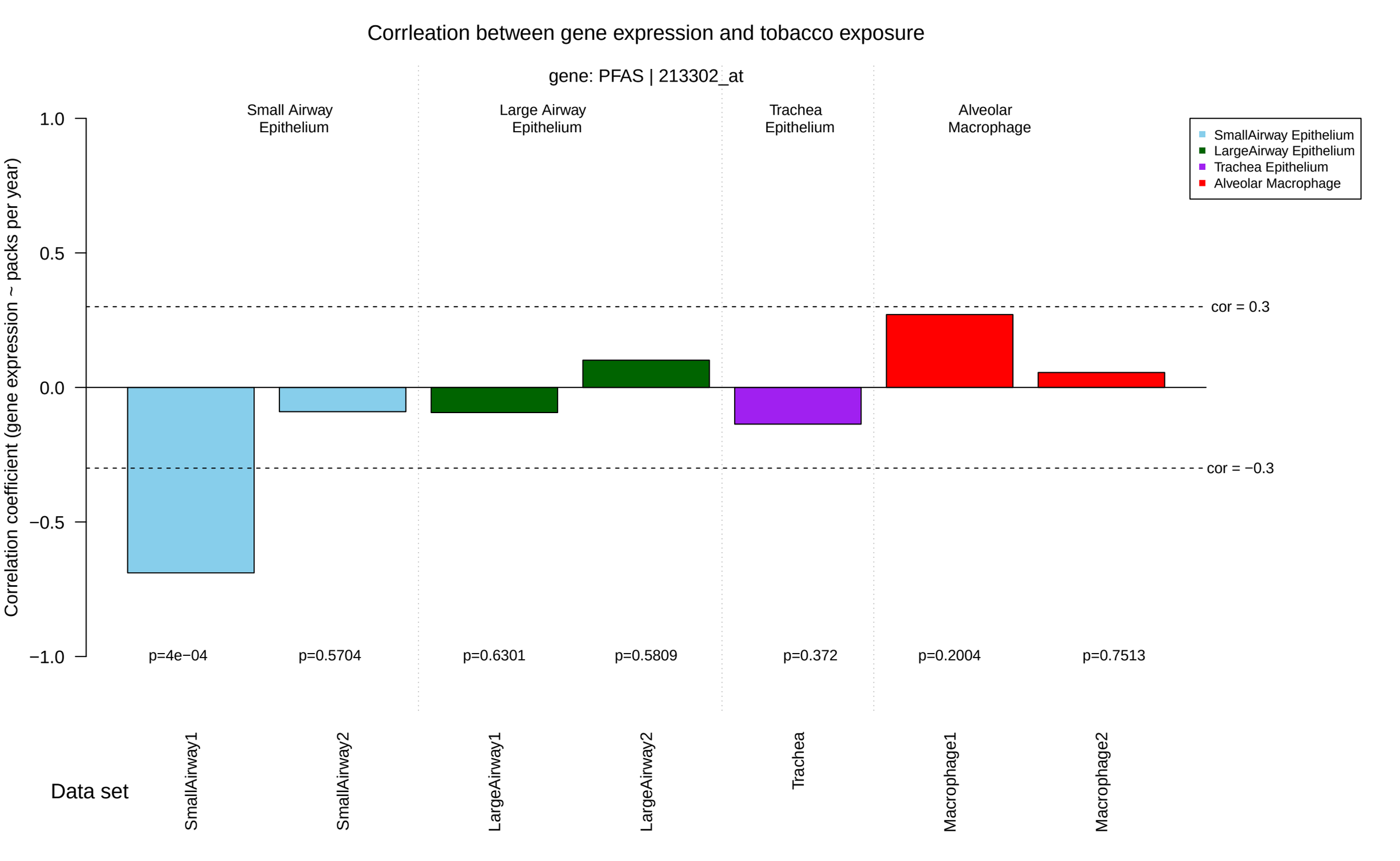

set: LargeAirway1 (p=0.0672)
PFAS | 213302_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7855)
PFAS | 213302_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1125)
PFAS | 213302_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6253)
PFAS | 213302_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4798)
PFAS | 213302_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PFAS | 213302_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=4e−04
p=0.5704
p=0.6301
p=0.5809
p=0.372
p=0.2004
p=0.7513
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2